

Program 10:

Demonstrate Kubernetes Dashboard and CLI for Cluster Monitoring

Step 1: Start Minikube with required resources

minikube start --cpus=2 --memory=4096

Step 2: Enable and open Kubernetes Dashboard

minikube dashboard

```
Dec 19 10:35 AM
isha@meow: ~/DevOps/prg10

isha@meow:~/DevOps/prg10$ minikube start --cpus=2 --memory=4096
🐳 minikube v1.37.0 on Ubuntu 24.04
🔧 Using the docker driver based on existing profile
❗ You cannot change the memory size for an existing minikube cluster. Please first delete the cluster.
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📦 Pulling base image v0.0.48 ...
🔄 Updating the running docker "minikube" container ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: storage-provisioner, default-storageclass
👏 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
isha@meow:~/DevOps/prg10$ minikube dashboard
🔧 Enabling dashboard ...
   ▪ Using image docker.io/kubernetesui/dashboard:v2.7.0
   ▪ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🔍 Verifying dashboard health ...
🚀 Launching proxy ...
🔍 Verifying proxy health ...
🔗 Opening http://127.0.0.1:33755/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser...
Gtk-Message: 10:28:22.531: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
```

Step 3: Create project directory

mkdir Program-10

cd Program-10

Step 4: Create Deployment YAML file

```
GNU nano 7.2 deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 5: Create Service YAML file

```
GNU nano 7.2 service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 6: Apply Deployment and Service

kubectl apply -f deployment.yaml

kubectl apply -f service.yaml

```
Dec 19 10:37 AM
isha@meow: ~/DevOps/prg10

isha@meow:~/DevOps/prg10$ nano deployment.yaml
isha@meow:~/DevOps/prg10$ nano service.yaml
isha@meow:~/DevOps/prg10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
isha@meow:~/DevOps/prg10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 7: Check running Pods

kubectl get pods

```
isha@meow:~/DevOps/prg10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-86rj6   1/1     Running   0           17s
nginx-deployment-6f9664446b-qltvf   1/1     Running   0           17s
```

Step 8: Check Kubernetes Nodes

kubectl get nodes

Step 9: Check Services

kubectl get svc

```
isha@meow:~/DevOps/prg10$ kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready    control-plane  17d   v1.34.0

isha@meow:~/DevOps/prg10$ kubectl get svc
NAME            TYPE          CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
kubernetes      ClusterIP     10.96.0.1        <none>       443/TCP          7d
nginx-service   NodePort      10.106.100.149   <none>       80:32681/TCP     21s
```

Step 10: View Pod Logs

kubectl logs nginx-deployment-6f9664446b-86rj6

```
iisha@meow:~/DevOps/prg10$ kubectl logs nginx-deployment-6f9664446b-86rj6
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:59:35 [notice] 1#1: using the "epoll" event method
2025/12/19 04:59:35 [notice] 1#1: nginx/1.29.4
2025/12/19 04:59:35 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:59:35 [notice] 1#1: OS: Linux 6.14.0-36-generic
2025/12/19 04:59:35 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:59:35 [notice] 1#1: start worker processes
2025/12/19 04:59:35 [notice] 1#1: start worker process 30
2025/12/19 04:59:35 [notice] 1#1: start worker process 31
2025/12/19 04:59:35 [notice] 1#1: start worker process 32
2025/12/19 04:59:35 [notice] 1#1: start worker process 33
2025/12/19 04:59:35 [notice] 1#1: start worker process 34
2025/12/19 04:59:35 [notice] 1#1: start worker process 35
2025/12/19 04:59:35 [notice] 1#1: start worker process 36
2025/12/19 04:59:35 [notice] 1#1: start worker process 37
```

Step 11: Describe Pod Details

kubectl describe pod nginx-deployment-6f9664446b-86rj6

```
iisha@meow:~/DevOps/prg10$ kubectl describe pod nginx-deployment-6f9664446b-86rj6Name:          nginx-deployment-6f9664446b-86rj6
Namespace:    default
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Fri, 19 Dec 2025 10:29:28 +0530
Labels:       app=nginx
              pod-template-hash=6f9664446b
Annotations:  <none>
Status:       Running
IP:           10.244.0.34
IPs:          IP: 10.244.0.34
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://424b33da3bfc66dbedd513f36da97e4eae90520d3729fd49a5ef849aec29cae8
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:          80/TCP
    Host Port:     0/TCP
    State:         Running
      Started:     Fri, 19 Dec 2025 10:29:35 +0530
    Ready:         True
    Restart Count:  0
    Environment:   <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-2rfwx (ro)
```

Step 12: Access Application via Minikube

minikube service nginx-service

```
isha@meow:~/DevOps/prg10$ minikube service nginx-service
```

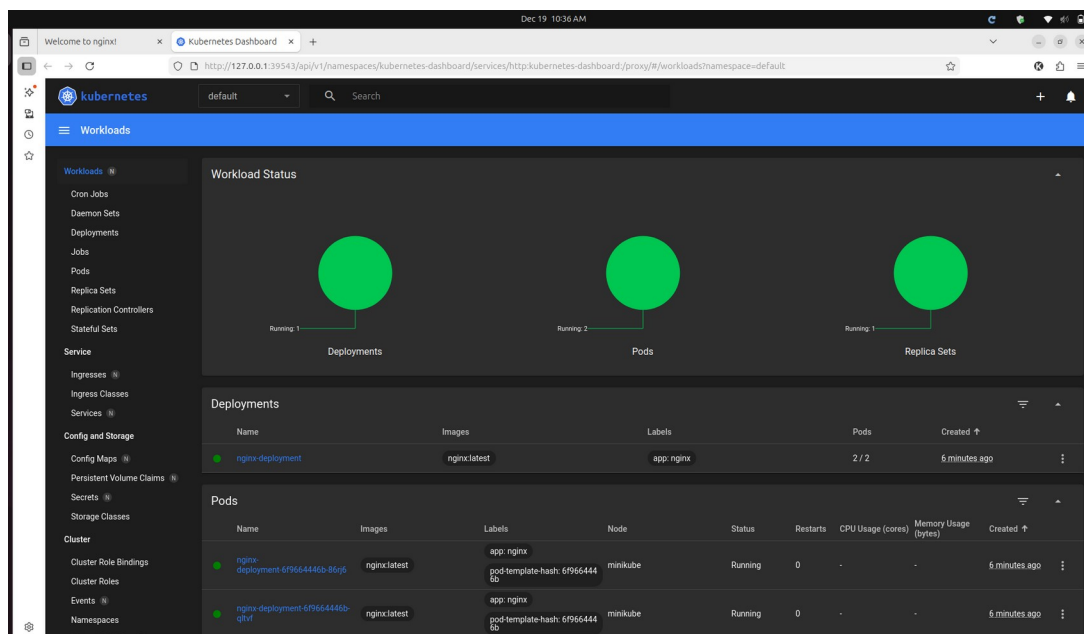
NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:32681

🌈 Opening service default/nginx-service in default browser...

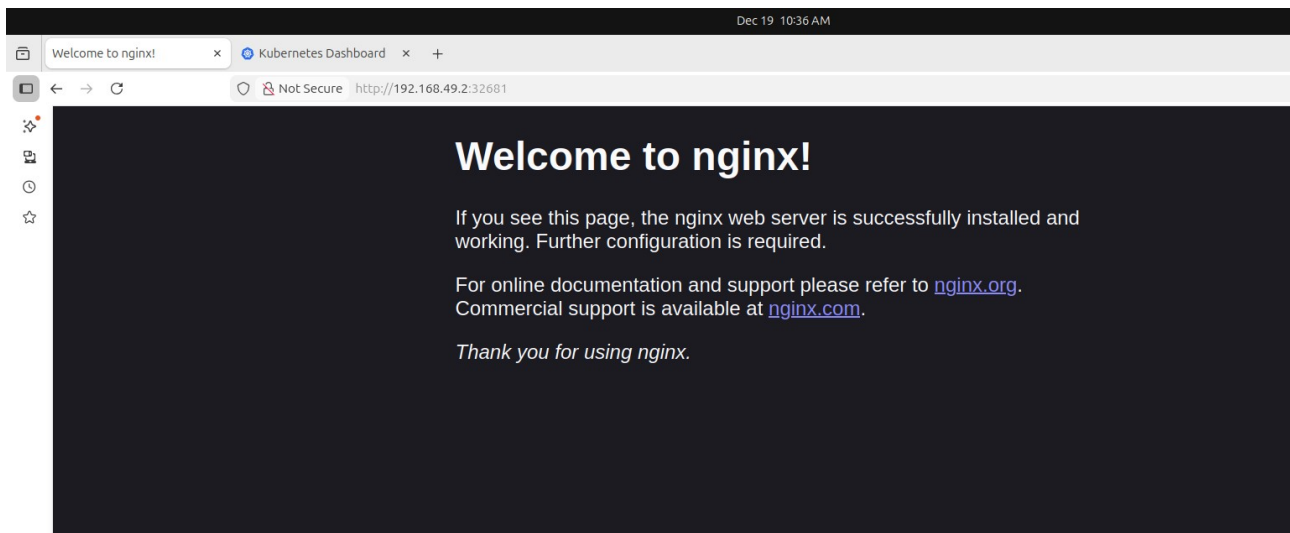
Step 13: Monitor Cluster using Dashboard

Action:

- Open Kubernetes Dashboard
- Observe pods, deployments, services, and logs



Deployments					
Name	Images	Labels	Pods	Created ↑	
nginx-deployment	nginx:latest	app: nginx	2 / 2	6 minutes ago	
Pods					
Name	Images	Labels	Node	Status	Restarts
nginx-deployment-6f966444b-86rj6	nginx:latest	app: nginx pod-template-hash: 6f966444b	minikube	Running	0
nginx-deployment-6f966444b-qitvf	nginx:latest	app: nginx pod-template-hash: 6f966444b	minikube	Running	0
Replica Sets					
Name	Images	Labels	Pods	Created ↑	
nginx-deployment-6f966444b	nginx:latest	app: nginx pod-template-hash: 6f966444b	2 / 2	6 minutes ago	



Step 14: Cleanup Resources

kubectl delete pods --all